Stage 1: API-Grounded Generation

Tasks

DeepEval Benchmark

API Knowledge Retrieval

LLM

Code Generation

Draft Code

Stage 2: Failure Diagnosis

Tensor Shape Tracing

Execution Failure

Traceback Text

Static Analysis

+

Dynamic Analysis

Root-Cause API Localization and Buggy Parameters Masking

infeatures=<mask>

Parameter-Masked Code

Stage 3: Shape-Aware Repair

Tensor Shape Annotation

[1,512,28,28]

[1,401408]

infeatures=<mask>

Shape-Annotated Code

Shape-Related Knowledge Retrieval

LLM

Shape-Aware Repair

infeatures=401408

Repaired Code

Supporting Resources

HuggingFace/Transformers

Locally Deployed LLMs

TensorFlow PyTorch PaddlePaddle

DL API Knowledge Base